Unit-7 (Association Rule Mining)

BScCSIT 8th Semester

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Outline

- **Frequent Itemset**
- **Mining Association Rules in Large Databases**
- **Association Rule Mining**
- **Application of Association Rule Mining**
- **Problems for association rule mining**
- **Apriori Algorithm**

Definition: Frequent Itemset

- **Itemset**
  A collection of one or more items
  - Example: {Milk, Bread, Sugar}
- **k-itemset**
  - An itemset that contains k items
- **Support count (P)**
  - Frequency of occurrence of an itemset
  - E.g. P(Bread,Milk,Sugar}) = 2
- **Support**
  - Fraction of transactions that contain an itemset
  - E.g. s({Bread, Milk, Sugar}) = 2/5
- **Frequent Itemset**
  - An itemset is called frequent if its support is equal or greater than an agreed upon **minimal value the support threshold**

TID	Items
1	Bread, Milk
2	Bread, coffee, eggs, sugar
3	Milk, coffee, coke, sugar
4	Bread, coffee, milk, sugar
5	Bread, coke , milk, sugar

Mining Association Rules in Large Databases

Introduction

- Association rule mining finds interesting association or correlation relationships among a large set of data items.
- With massive amounts of data continuously being collected and stored , many industries are becoming interested in mining association huge amounts of business transaction records which helps in many business decision making processes, such as **catalog design**, **cross-marketing**, and **loss-leader analysis.**

Mining Association Rules in Large Databases

- A typical example of association rule mining is **market basket analysis**.
- This process analyzes customer buying habits by finding associations between the different items that customers place in their “shopping baskets”.
- The discovery of such associations can help retailers develop marketing strategies by gaining insight into which items are frequently purchased together by customers.
- For instance, if customers are buying milk, how likely are they to also buy bread(and what kind of bread)on the same trip to the supermarket? Such information can lead to increased sales by helping retailers do selective marketing and plan their shelf space. For example, placing milk And bread within close proximity may further encourage the sale of these items together within single visits to the store.
- **How can we find association rules from large amounts of data, where the data are either transactional or relational? Which association rules are the most interesting ?How can we help or guide the mining procedure to discover interesting associations? What language constructs are useful in defining a data mining query language for association rule mining.**

Association Rule

- Association Rule Mining was first introduced in 1993 Proposed by Agrawal et al.

Definition:

- Let I ={I1, I2, … , Im} be a set of m distinct attributes, also called **literals**. Let D be a database, where each record (tuple) T has a unique identifier, and contains a set of items such that **T⊆I**. **An association rule** is an implication of the form **X⇒Y,** where X, Y⊂I, are sets of items called itemsets, and X ∩Y=φ. Here, X is called **antecedent**, and Y **consequent**.
- Two important **Rule strength measures** for association rules,support (s) and confidence (α), can be defined.

Support (s)

- The **support (s)** of an association rule is the ratio (in percent) of the records that contain $X \cup Y$ to the total number of records in the database.
  - **support (X=>Y) = support(XUY) or *sup* = Pr($X \cup Y$)**
- Therefore, if we say that the support of a rule is 5% then it means that 5% of the total records contain $X \cup Y$.
- Support is the statistical significance of an association rule.
- **Grocery store** managers probably would not be concerned about how peanut butter and bread are related if less than 5% of store transactions have this combination of purchases.
- While a high support is often desirable for association rules, this is not always the case.
- **For example**, if we were using association rules to predict the failure of telecommunications switching nodes based on what set of events occur prior to failure, even if these events do not occur very frequently association rules showing this relationship would still be important.

Confidence (α)

- For a given number of records, **confidence (α)** is the ratio (in percent) of the number of records that contain $\mathbf{X} \cup \mathbf{Y}$ to the number of records that contain X.
  - **confidence(X=>Y) = support(XUY) / support(X)**

    or
  - ***conf* = Pr(*Y* | *X*)**
- Thus, if we say that a rule has a confidence of 85%, it means that 85% of the records containing X also contain Y.
- The confidence of a rule indicates the degree of correlation in the dataset between X and Y.
- Confidence is a measure of a rule's strength. Often a large confidence is required for association rules.

Support and Confidence

- *support* of X in D is $count(X)/|D|$
- For an association rule $X \Rightarrow Y$, we can calculate

  support $(X \Rightarrow Y)$ = support (XY)

  confidence $(X \Rightarrow Y)$ = support (XY)/support (X)
- Relate Support (S) and Confidence (C) to Joint and Conditional probabilities
- There could be exponentially many A-rules
- Interesting association rules are (for now) those whose S and C are greater than minSup and minConf (some thresholds set by data miners)

Support and Confidence

Support count:

The support count of an itemset *X*, denoted by *X.count*, in a data set *T* is the number of transactions in *T* that contain *X*. Assume *T* has *n* transactions. Then,

$$support = \frac{(X \cup Y).count}{n}$$

$$confidence = \frac{(X \cup Y).count}{X.count}$$

Example of support Measure

Table 1 Example of Support Measure.

TID	Items	Support = Occurence / Total Support
1	ABC	
2	ABD	Total Support = 5
3	BC	Support {AB} = 2 / 5 = 40%
4	AC	Support {BC} = 3 / 5 = 60%
5	BCD	Support {ABC} = 1 / 5 = 20%

Example of Confidence Measure

Table 2 Example of Confidence Measure.

TID	Items	
		Given $X \Rightarrow Y$
1	ABC	Confidence = Occurrence {Y} / Occurrence {X}
2	ABD	
3	BC	Confidence $\{A \Rightarrow B\} = 2/3 = 66\%$
4	AC	Confidence $\{B \Rightarrow C\} = 3/4 = 75\%$
5	BCD	Confidence $\{AB \Rightarrow C\} = 1/2 = 50\%$

Association Rules

Confidence

- We should only consider rules derived from itemsets with high support, and that also have high confidence.
- “A rule with low confidence is not meaningful.”
- Rules don’t explain anything, they just point out hard facts in data volumes.

Example

Example: Database with transactions (customer_# : item_a1, item_a2, ...)

1: 3, 5, 8.
2: 2, 6, 8.
3: 1, 4, 7, 10.
4: 3, 8, 10.
5: 2, 5, 8.
6: 1, 5, 6.
7: 4, 5, 6, 8.
8: 2, 3, 4.
9: 1, 5, 7, 8.
10: 3, 8, 9, 10.

Conf ({5} => {8}) ?

supp({5}) = 5 , supp({8}) = 7 , supp({5,8}) = 4,

then **conf({5} => {8}) = 4/5 = 0.8 or 80%**

Example

Example: Database with transactions (customer_# : item_a1, item_a2, ...)

1: 3, 5, 8.
2: 2, 6, 8.
3: 1, 4, 7, 10.
4: 3, 8, 10.
5: 2, 5, 8.
6: 1, 5, 6.
7: 4, 5, 6, 8.
8: 2, 3, 4.
9: 1, 5, 7, 8.
10: 3, 8, 9, 10.

Conf ({5} => {8}) ? 80% Done. Conf ({8} => {5}) ?

supp({5}) = 5 , supp({8}) = 7 , supp({5,8}) = 4,

then **conf({8} => {5}) = 4/7 = 0.57 or 57%**

Example

Example: Database with transactions (customer_# : item_a1, item_a2, ...)

Conf ({5} => {8}) ? 80% Done.

Conf ({8} => {5}) ? 57% Done.

Rule ({5} => {8}) more meaningful then

Rule ({8} => {5})

Example

Example: Database with transactions (customer_# : item_a1, item_a2, ...)

1:	3, 5, 8.
2:	2, 6, 8.
3:	1, 4, 7, 10.
4:	3, 8, 10.
5:	2, 5, 8.
6:	1, 5, 6.
7:	4, 5, 6, 8.
8:	2, 3, 4.
9:	1, 5, 7, 8.
10:	3, 8, 9, 10.

Conf ({9} => {3}) ?

supp({9}) = 1 , supp({3}) = 4 , supp({3,9}) = 1,

then **conf({9} => {3}) = 1/1 = 1.0 or 100%. OK?**

Example

Example: Database with transactions (customer_# : item_a1, item_a2, ...)

Conf({9} => {3}) = 100%. Done.

Notice: High Confidence, Low Support.

-> Rule ({9} => {3}) not meaningful

Support, Confidence and Lift

- Association rule is of the form $X \Rightarrow Y$, where $X, Y \subset I$ are itemsets and $X \cap Y = \emptyset$.
- support$(X \Rightarrow Y) = P(X \cup Y) = \text{count}(X \cup Y)/\text{dbsize}$.
- confidence$(X \Rightarrow Y) = P(Y|X) = \text{count}(X \cup Y)/\text{count}(X)$.
- Therefore, always support$(X \Rightarrow Y) \leq$ confidence$(X \Rightarrow Y)$.
- Typical values for min_sup in practical applications from 1 to 5%, for min_conf more than 50%.
- lift$(X \Rightarrow Y) = P(Y|X)/P(Y)$

  $= \text{count}(X \cup Y)*\text{dbsize} / \text{count}(X)*\text{count}(Y)$,

  measures the increase in likelihood of Y given X vs. random (= no info).

Apriori Algorithm

- Apriori is a seminal algorithm proposed by R. Agrawal and R. Srikant in 1994 for mining frequent itemsets for Boolean association rules.
- The name of the algorithm is based on the fact that the algorithm uses prior knowledge of frequent itemset properties.
- The Apriori algorithm is the most well known association rule algorithm and is used in most commercial products.
- **Two steps**:
  - Find all itemsets that have minimum support (*frequent itemsets*, also called large itemsets).
  - Use frequent itemsets to generate rules.

The Apriori Algorithm

- Pseudo-code:

  C_k: Candidate itemset of size k

  L_k : frequent itemset of size k

  L_1 = {frequent items};

  for (k = 1; L_k !=∅; k++) **do begin**

  C_{k+1} = candidates generated from L_k;

  Important! How?! Next slide…

  for each transaction t in database do

  increment the count of all candidates in C_{k+1} that are contained in t

  L_{k+1} = candidates in C_{k+1} with min_support

  end

  return $\cup_k L_k$;

Important Details of Apriori

- How to generate candidates?
  - Step 1: self-joining L_k
  - Step 2: pruning
- Example of candidate-generation
  - L_3={*abc, abd, acd, ace, bcd*}
  - Self-joining: $L_3 * L_3$
    - *abcd* from *abc* and *abd*
    - *acde* from *acd* and *ace*
    - Not *abcd* from *abd* and *bcd* !

      This allows efficient implementation: sort candidates L_k lexicographically to bring together those with identical (k-1)-prefixes, …
  - Pruning:
    - *acde* is removed because *ade* is not in L_3
  - C_4={*abcd*}

How to Generate Candidates?

- Suppose the items in L_{k-1} are listed in an order
- Step 1: self-joining L_{k-1}

  insert into $\boldsymbol{C_k}$

  select $\boldsymbol{p.item_1, p.item_2, \ldots, p.item_{k-1}, q.item_{k-1}}$

  from $\boldsymbol{p \in L_{k-1}, q \in L_{k-1}}$

  where $\boldsymbol{p.item_1 = q.item_1, \ldots, p.item_{k-2} = q.item_{k-2}, p.item_{k-1} < q.item_{k-1}}$
- Step 2: pruning

  forall ***itemsets c in*** $\boldsymbol{C_k}$ do

  forall ***(k-1)-subsets s of c*** do

  if *(s is not in* L_{k-1}*)* **then delete** *c* **from** C_k

Requirements for effective candidate generation procedure

1. It should avoid generating too many unnecessary candidates. A candidate itemset is unnecessary if at least one of its subsets is infrequent. Such a candidate is guaranteed to be infrequent according to the anti-monotone property of support.

2. It must ensure that the candidate set is complete, i.e., no frequent itemsets are left out by the candidate generation procedure. To ensure completeness, the set of candidate itemsets must subsume the set of all frequent itemsets, i.e., $\forall k : F_k \subseteq C_k$.

3. It should not generate the same candidate itemset more than once. For example, the candidate itemset $\{a, b, c, d\}$ can be generated in many ways—by merging $\{a, b, c\}$ with $\{d\}$, $\{b, d\}$ with $\{a, c\}$, $\{c\}$ with $\{a, b, d\}$, etc. Generation of duplicate candidates leads to wasted computations and thus should be avoided for efficiency reasons.

The Apriori Algorithm: Example

TID	List of Items
T100	I1, I2, I5
T100	I2, I4
T100	I2, I3
T100	I1, I2, I4
T100	I1, I3
T100	I2, I3
T100	I1, I3
T100	I1, I2 ,I3, I5
T100	I1, I2, I3

- Consider a database, D , consisting of 9 transactions.
- Suppose min. support count required is 2 (i.e. min_sup = 2/9 = 22 %)
- Let minimum confidence required is 70%.
- We have to first find out the frequent itemset using Apriori algorithm.
- Then, Association rules will be generated using min. support & min. confidence.

Step 1: Generating 1-itemset Frequent Pattern

Scan D for count of each candidate

Itemset	Sup.Count
{I1}	6
{I2}	7
{I3}	6
{I4}	2
{I5}	2

C_1

Compare candidate support count with minimum support count

Itemset	Sup.Count
{I1}	6
{I2}	7
{I3}	6
{I4}	2
{I5}	2

L_1

- The set of frequent 1-itemsets, L_1 , consists of the candidate 1-itemsets satisfying minimum support.
- In the first iteration of the algorithm, each item is a member of the set of candidate.

Step 2: Generating 2-itemset Frequent Pattern

Generate C_2 candidates from L_1

Itemset
{I1, I2}
{I1, I3}
{I1, I4}
{I1, I5}
{I2, I3}
{I2, I4}
{I2, I5}
{I3, I4}
{I3, I5}
{I4, I5}

C_2

Scan D for count of each candidate

Itemset	Sup. Count
{I1, I2}	4
{I1, I3}	4
{I1, I4}	1
{I1, I5}	2
{I2, I3}	4
{I2, I4}	2
{I2, I5}	2
{I3, I4}	0
{I3, I5}	1
{I4, I5}	0

C_2

Compare candidate support count with minimum support count

Itemset	Sup Count
{I1, I2}	4
{I1, I3}	4
{I1, I5}	2
{I2, I3}	4
{I2, I4}	2
{I2, I5}	2

L_2

Step 2: Generating 2-itemset Frequent Pattern

- To discover the set of frequent 2-itemsets, L_2 , the algorithm uses L_1 *Join* L_1 to generate a candidate set of 2-itemsets, C_2.
- Next, the transactions in D are scanned and the support count for each candidate itemset in C_2 is accumulated (as shown in the middle table).
- The set of frequent 2-itemsets, L_2 , is then determined, consisting of those candidate 2-itemsets in C_2 having minimum support.
- Note: We haven't used Apriori Property yet.

Step 3: Generating 3-itemset Frequent Pattern

Scan D for count of each candidate

Itemset
{I1, I2, I3}
{I1, I2, I5}

C_3

Scan D for count of each candidate

Itemset	Sup. Count
{I1, I2, I3}	2
{I1, I2, I5}	2

C_3

Compare candidate support count with min support count

Itemset	Sup Count
{I1, I2, I3}	2
{I1, I2, I5}	2

L_3

- The generation of the set of candidate 3-itemsets, C_3 , involves use of the Apriori Property.
- In order to find C_3, we compute L_2 *Join* L_2.
- C_3 = L2 *Join* L2 = {{I1, I2, I3}, {I1, I2, I5}, {I1, I3, I5}, {I2, I3, I4}, {I2, I3, I5}, {I2, I4, I5}}.
- Now, Join step is complete and Prune step will be used to reduce the size of C_3. Prune step helps to avoid heavy computation due to large C_k.

Step 3: Generating 3-itemset Frequent Pattern

- Based on the Apriori property that all subsets of a frequent itemset must also be frequent, we can determine that four latter candidates cannot possibly be frequent. How ?
- For example , lets take {I1, I2, I3}. The 2-item subsets of it are {I1, I2}, {I1, I3} & {I2, I3}. Since all 2-item subsets of {I1, I2, I3} are members of L_2, We will keep {I1, I2, I3} in C_3.
- Lets take another example of {I2, I3, I5} which shows how the pruning is performed. The 2-item subsets are {I2, I3}, {I2, I5} & {I3,I5}.
- BUT, {I3, I5} is not a member of L_2 and hence it is not frequent violating Apriori Property. Thus We will have to remove {I2, I3, I5} from C_3.
- Therefore, C_3 = {{I1, I2, I3}, {I1, I2, I5}} after checking for all members of result of Join operation for Pruning.
- Now, the transactions in D are scanned in order to determine L_3, consisting of those candidates 3-itemsets in C_3 having minimum support.

Step 4: Generating 4-itemset Frequent Pattern

- The algorithm uses L_3 *Join* L_3 to generate a candidate set of 4-itemsets, C_4. Although the join results in {{I1, I2, I3, I5}}, this itemset is pruned since its subset {{I2, I3, I5}} is not frequent.
- Thus, $C_4 = \varphi$, and algorithm terminates, having found all of the frequent items. This completes our Apriori Algorithm.
- What's Next ?

  These frequent itemsets will be used to generate strong association rules (where strong association rules satisfy both minimum support & minimum confidence).

Step 5: Generating Association Rules from Frequent Itemsets

- Procedure:
  - For each frequent itemset ***"l"***, generate all nonempty subsets of ***l***.
  - For every nonempty subset ***s*** of ***l***, output the rule **"s → (l-s)"** if **support_count(l) / support_count(s) >= min_conf** where min_conf is minimum confidence threshold.

- Back To Example:

We had L = {{I1}, {I2}, {I3}, {I4}, {I5}, {I1,I2}, {I1,I3}, {I1,I5}, {I2,I3}, {I2,I4}, {I2,I5}, {I1,I2,I3}, {I1,I2,I5}}.

  – Lets take ***l*** = {I1,I2,I5}.
  – Its all nonempty subsets are {I1,I2}, {I1,I5}, {I2,I5}, {I1}, {I2}, {I5}.

Step 5: Generating Association Rules from Frequent Itemsets

- Let minimum confidence threshold is , say 70%.
- The resulting association rules are shown below, each listed with its confidence.
  - R1: I1 ^ I2 → I5
    - Confidence = sc{I1,I2,I5}/sc{I1,I2} = 2/4 = 50%
    - R1 is Rejected.
  - R2: I1 ^ I5 → I2
    - Confidence = sc{I1,I2,I5}/sc{I1,I5} = 2/2 = 100%
    - R2 is Selected.
  - R3: I2 ^ I5 → I1
    - Confidence = sc{I1,I2,I5}/sc{I2,I5} = 2/2 = 100%
    - R3 is Selected.

Step 5: Generating Association Rules from Frequent Itemsets

- R4: I1 → I2 ^ I5
  - Confidence = sc{I1,I2,I5}/sc{I1} = 2/6 = 33%
  - R4 is Rejected.
- R5: I2 → I1 ^ I5
  - Confidence = sc{I1,I2,I5}/{I2} = 2/7 = 29%
  - R5 is Rejected.
- R6: I5 → I1 ^ I2
  - Confidence = sc{I1,I2,I5}/ {I5} = 2/2 = 100%
  - R6 is Selected.

In this way, We have found three strong association rules.

Other Example

The Apriori Algorithm—An Example

min_sup = 2

Database TDB

Tid	Items
10	A, C, D
20	B, C, E
30	A, B, C, E
40	B, E

1st scan →

C_1

Itemset	sup
{A}	2
{B}	3
{C}	3
{D}	1
{E}	3

→ L_1

Itemset	sup
{A}	2
{B}	3
{C}	3
{E}	3

C_2

Itemset
{A, B}
{A, C}
{A, E}
{B, C}
{B, E}
{C, E}

2nd scan →

C_2

Itemset	sup
{A, B}	1
{A, C}	2
{A, E}	1
{B, C}	2
{B, E}	3
{C, E}	2

→ L_2

Itemset	sup
{A, C}	2
{B, C}	2
{B, E}	3
{C, E}	2

C_3

Itemset
{B, C, E}

3rd scan →

L_3

Itemset	sup
{B, C, E}	2

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Application of Association Rule Mining

- Different application areas of association rule mining are:
- **Market Basket Analysis**
- **Medical diagnosis**
- **Protein Sequences**
- **Census data**
- **CRM of credit card business**

Market Basket Analysis

- A broadly-used example of association rule mining is **market basket analysis**.
- In market basket databases consist of a large no. of records and in each record all items bought by a customer on a single purchase transaction are listed.
- Managers would be paying attention to know that which groups of items are constantly purchased together.
- This data is used by them to adjust store layouts (placing items optimally with respect to each other), to cross-sell, to promotions, to catalog design and to identify customer segments based on buying patterns.
- For example, suppose a shop database has 200,000 point-of-sale transactions, out of which 4,0000 include both items A and B and 1600 of these include item C, the association rule "If A and B are purchased then C is purchased on the same trip" has a support of 1600 transactions (alternatively 0.8% = 1600/200,000) and a confidence of 4% (=1600/4,0000).

Market Basket Analysis

- A large amount of sales data is produced by the software supporting these barcode based purchasing/ordering system which is typically captured in “baskets”.
- Commercial organizations are interested in discovering “association rules” that identify patterns of purchases, such that the presence of one item in a basket will imply the presence of one or more additional items.
- This “market basket analysis” result can be used to suggest combinations of products for special promotions or sales.

Medical diagnosis

- Association rules can be used in medical analysis for assisting physicians to cure patients.
- The common problem of the induction of reliable analytic rules is hard as theoretically no induction process by itself can guarantee the accuracy of induced hypotheses.
- Basically diagnosis is not an easy process because of unreliable diagnosis tests and the presence of noise in training examples.
- This may result in hypotheses with insufficient prediction correctness which is too unreliable for critical medical applications

Census data

- Censuses make a huge variety of general statistical information on society available to both researchers and the general community .
- The information related to population and economic census can be forecasted in planning public services(education, health, transport, funds) as well as in public business(for setup new factories, shopping malls or banks and even marketing particular products).
- The application of data mining techniques to census data and more generally to official data has great potential in supporting good community policy and in underpinning the effective functioning of a democratic society.

CRM of credit card business

- Customer Relationship Management (CRM), through which, banks expect to identify the preference of different customer groups, products and services adapted to their liking to enhance the cohesion between credit card customers and the bank, has become a topic of great interest .
- The collective application of association rule techniques reinforces the knowledge management process and allows marketing personnel to know their customers well to provide better quality services.

Potential problems for association rule mining

1. To make frequent pattern mining an essential task in data mining, much research is needed to further develop pattern based mining methods.
2. Mechanisms for deep understanding and interpretation of patterns, e.g., semantic annotation for frequent patterns, and contextual analysis of frequent patterns are needed.
3. Most approaches are based on some strict assumptions. We should generalize these algorithms so that they can be more widely used in practice.
4. We should explore more effective algorithms and look for a balance between disclosure cost, computation cost and communication cost.
5. More efficient and scalable methods for Association Rule mining should be developed.
6. Single scan and online mining methods should be developed to discover knowledge or patterns from data streams.

Potential problems for association rule mining

7. Database-independent measurements should be established
8. Efficient and effective hidden pattern mining methods and systems should be developed.
9. Deep-level association rules should be identified.
10. Techniques for mining association rules in multi-databases should be explored.
11. Effective techniques for Web Usage Mining should be developed
12. Association rule mining should be applied to social network analysis and mining for business
13. Frequent pattern mining has started playing an important role in software bug detection and analysis. Many interesting methods have been developed to trace crashing bugs. But more efficient methods are still needed.

Methods to Improve Apriori's Efficiency

- Hash-based itemset counting: A k-itemset whose corresponding hashing bucket count is below the threshold cannot be frequent.
- Transaction reduction: A transaction that does not contain any frequent k-itemset is useless in subsequent scans.
- Partitioning: Any itemset that is potentially frequent in DB must be frequent in at least one of the partitions of DB.
- Sampling: mining on a subset of given data, lower support threshold + a method to determine the completeness.
- Dynamic itemset counting: add new candidate itemsets only when all of their subsets are estimated to be frequent.

Thank you !!!